

DOUBLE PENDULUM EXPERIMENT

Goal: acquire reliable video data from two nearly identical releases, reconstruct the canonical state of the double pendulum, and test whether the trajectories show an interval of approximately exponential separation.

Stage	Main task
1. Prepare	Build and fully check the pendulum, measure all physical parameters, and calibrate the camera.
2. Record	Make videos with nearly identical initial conditions and fixed recording settings.
3. Track	Extract marker coordinates, inspect tracking quality, and export canonical-coordinate CSV files.
4. Compare	Process each file separately, align elapsed time, calculate phase-space separation, and fit the early growth region.
5. Evaluate	Evaluate the data and the analysis critically before drawing conclusions.
6. Report	Write a report that demonstrates that you have successfully performed the experiment, processed the data and drawn critical conclusions.

1. Learning outcomes

Upon successful completion of this experiment, the students will be able to:

1. **Explain and apply** the physical theory of the double pendulum to formulate a feasible experimental research question about sensitivity to initial conditions.
2. **Conduct** a reproducible experiment, including measurement planning, calibration, control of recording conditions, and estimation of relevant measurement uncertainties.
3. **Develop and validate** a video-tracking and data-processing workflow that reconstructs the generalized coordinates, angular velocities, generalized momenta, and phase-space trajectories from calibrated image data.
4. **Analyse** the divergence of nearby trajectories using an appropriately normalized phase-space distance and a finite-time divergence rate, distinguishing genuine dynamical behaviour from noise, dissipation, calibration errors, tracking errors, and synchronization errors.
5. **Communicate and critically discuss** the investigation in a clear scientific report that presents the research question, theory, methods, visualizations, results, and possible improvements, while distinguishing observations from interpretations.
6. **Justify** the methodological choices made throughout the experiment and **evaluate** their influence on the robustness of the conclusions.

2. Apparatus preparation and calibration

1. Build the double pendulum and check it fully: rigid support, secure pivots, free motion, no collisions, and approximately planar motion.
2. Measure and record the effective arm lengths L_1 and L_2 , masses m_1 and m_2 , and their uncertainties.
3. Mount the camera rigidly, approximately perpendicular to the motion plane. Keep the whole swing visible.
4. Place a ruler or calibration target in the same plane as the pendulum and determine pixels per centimetre.
5. Record or export a first-frame image at the original pixel resolution. Mark the fixed pivot and calibration endpoints on a copy.

Quality pointer: Perspective error is systematic. A ruler several centimetres in front of or behind the pendulum does not provide a valid calibration for the pendulum plane.

Minimum parameter record

Quantity	Value	Uncertainty	Method
L_1			
L_2			
m_1			
m_2			
pixels per cm			
pivot x and y [pixels]			
frame rate			

3. Video acquisition

6. Use a fixed camera, stable lighting, short exposure where possible, and a frame rate high enough to avoid large marker jumps between frames. A modern smart phone should be able to do the job.
7. Use compact, high-contrast markers. Tracking both moving joints is preferred.
8. Record multiple releases with a tiny difference in initial condition. Keep apparatus, calibration, camera, and lighting unchanged.
9. Do not worry if a hand covers the first part of each video. Each recording can be processed later.
10. Immediately inspect several frames from slow and fast parts of the motion for blur, out-of-plane motion, and marker confusion.

Stop and repeat the recording if: the support moves, a marker leaves the frame, the motion becomes strongly out of plane, or the tracker cannot distinguish the marker from the background.

Goals for Day 1:

Before leaving the lab on the first day, make sure that you have a setup that works and that you can gather meaningful data. How do you verify that the data is meaningful?

Goals for Day 2:

With the initial data from day 1, make sure that you have a first analysis code that can extract the coordinates of the double pendulum. For day 2, you need to gather more data with different initial conditions or potentially by making changes to the setup. Making a detailed plan before the session. By the end of Day 2, all data should be collected.

4. Tracking and coordinate reconstruction

11. Track marker frame by frame using python (see example notebook on Brightspace)
12. Inspect the original frame and annotated frame together. Do not judge tracking from coordinates alone.
13. Convert pixels to metres using the measured calibration and subtract the fixed-pivot position.
14. Reconstruct the absolute angles θ_1 and θ_2 (see attached theory note).
15. Plot $\phi_1(t)$ and $\phi_2(t)$ before differentiating.

5. Velocities and canonical coordinates

16. Calculate the angular velocity $\omega_1(t)$ and $\omega_2(t)$ from the measured angles, then calculate the generalized momenta $p_1(t)$ and $p_2(t)$ using the measured lengths and masses.
17. Export one canonical-coordinate CSV per recording

Quality pointer: Plot raw angles and derivatives and make sure that the curves look smooth and consistent. If the data at this point is not good enough, you need to go back and take better data.

6. Compare the recordings (start with a single pair of two recordings)

18. Choose a separate first usable time for each file (after hands or any tracking contamination have disappeared).
19. Crop each file independently and reset the first retained sample to elapsed time $\tau = 0$.
20. Check that the two cropped initial states are genuinely close in θ_1 , θ_2 , p_1 and p_2 .
21. Calculate a dimensionless phase-space distance using declared angular and momentum scales.

Recommended distance:

$$\delta(\tau) = \sqrt{\frac{\theta_1^2}{\theta_{1,norm}^2} + \frac{p_1^2}{p_{1,norm}^2} + \frac{\theta_2^2}{\theta_{2,norm}^2} + \frac{p_2^2}{p_{2,norm}^2}}$$

Coding decision: Angles and momenta have different units, so they must not be added in an unscaled Euclidean norm. Instead, normalize each one appropriately. Report every normalization scale you use.

7. Test for exponential separation

22. Plot $\delta(\tau)$ on linear and logarithmic vertical axes.
23. Identify an early interval where $\log \delta$ is approximately linear. Exclude the initial noise and the later saturation region.
24. Fit $\log \delta = \lambda \tau + b$ over the chosen interval and report λ , the fit interval, and the initial separation.
25. Repeat the fit for nearby intervals. A trustworthy conclusion should not depend on one very specific choice.

Interpretation: Describe λ as a finite-time divergence rate. An early rise followed by saturation is compatible with chaos, but noise, synchronization error, and dissipation must also be considered. How do you make your conclusion stronger?

8. Required quality checks

- Calibration: repeat endpoint selection and estimate the resulting length-scale uncertainty.
- Tracking: inspect masks and annotated frames at high speed and near crossings.
- Geometry: verify that the pivot is fixed and motion remains approximately planar.
- Alignment: vary each crop time slightly and examine whether the conclusion is stable.
- Normalization: vary phase-space scales modestly and check qualitative robustness.
- Components: compare angle-only separation with full canonical separation.

9. Laboratory report

The report should as a minimum contain:

- Apparatus image or diagram with pendulum arms, L1, L2, masses, and markers identified.
- Calibration image and a table of measured parameters with uncertainties.
- Example original frame, threshold mask, and annotated tracking frame.
- Canonical phase-space projections.
- Estimation of λ (the exponential separation constant) from data
- Discuss showing how λ changes with analysis choices.
- A conclusion that separates observed evidence from limitations.